Neuroling = brain + language theory + method | 3 Problem spaces: (RQ) Localization + LAD(y/n) (Method) Methodology: map L in brain + brain imaging (theo debate) Architecture/sys org: Modular or distributed Overview level = map not details (mechanism)
传统损伤研究（**lesion studies**）在脑区定位上存在系统性局限，难以精确推断功能-脑区映射关系。1️⃣Sampling bias（取样偏差）表面意思：某些脑区更容易受损（如血供脆弱区域）深层机制：中风（stroke）分布不是随机的 + 血管结构 → lesion 分布偏向特定区域 关键推论：我们研究到的“重要脑区”可能只是“更容易坏的脑区” 常见误区：❌ “某区域经常损伤 → 一定最重要” ✅ 实际可能是：sampling artifact（取样假象）2️⃣Lesion is too large（损伤范围过大）表面：损伤区域 > 实际关键区域 本质问题：lesion ≠ single functional unit 因为：大脑是 network lesion 通常影响：多个区域连接（white matter）高分理解：功能缺失可能来自“连接断裂”，而不是某个点本身👉 → 引出：disconnection theory 3️⃣Mapping is coarse（定位粗糙）问题：无法精确到：神经元群 micro-circuit 深层问题 传统方法是：lesion → behavior 但缺少：精细空间分辨率 实时动态信息 4️⃣Long-term tracking 问题：从受伤 → 行为变化 → 死后解剖
深层影响：数据延迟+ 样本量小+ 难以控制变量 ⚠️ 方法论问题：low reproducibility + low scalability 5️⃣Neural reorganization 表面：大脑会“补偿” 深层机制：neuroplasticity + functional reassignment 核心问题：你看到的行为 ≠ 原始功能定位 举例：A 区损伤 B 区接管功能 你可能误以为：B 原本负责这个功能 ❌ post-lesion brain ≠ normal brain
Early localization studies identified key language regions, but modern research shows that language emerges from interactions within distributed neural networks. Why did the field move from localization to network models? lesion limitations + connectivity importance + neuroimaging evidence
Production pathway: concept - prefrontal cortex: ideas generating/discourse planning（语篇建构）- chaos in thoughts/pragmatic deficit (speak in no org) ➡️Lexical selection - MTG: semantic competition (many words activated) - semantic paraphasia (semantic substitution: say wrong word)➡️Retrieval - MTG + angular gyrus - tip-of-the-tongue (TOT: know but cant say): phonological failure (short-term) - anomia: lexical failure (long-term)➡️Phonological encoding - Broca - speech errors - non-fluent/agrammatism➡️Articulation（发音）- motor cortex - make sounds - dysarthria (context right, sounds wrong) **Reading/comprehension pathway**: Visual input - occipital lobe - visual processing - cortical blindness (due to Primary visual cortex but not eyes)➡️Orthographic processing - VWFA in LFG/occipito-temporal cortex - reco shapes/forms - pure alexia (can only read char by char)➡️Lexical access - temporal cortex - reco faster on high freq - lexical access difficulty (many pauses/circumlocution: that one you know that/semantic substitution)➡️Semantic processing - Wernicke - N400 (semantic integration difficulty increase/N400 increase on words m mismatch in sent) - fluent but meaningless - comprehension deficit (see/hear yes but understand no/wrong)➡️Conceptual integration - temporal–frontal network - contextual comprehension (linguistic context: words ahead/behind + situational context: physical env + world knowledge (freq) + discourse context: whole org/logic) - comprehension deficit in discourse level
Lexical level(word) -> syntactic level(gram) ->semantic integration/context level (word -> m)-> discourse level (unders each sent yes whole conversation no) ｜Production + compre share representation (lexicon: lexical sys + Semantics: sem sys) with opp processing direction: is network-based + non-linear/parallel + feedback
phonology：phoneme（音位）对比性（contrastive）是 **categorical system**（离散系统），不是连续信号phonology = abstract sound system temporal ｜**Morphology**（形态）词的内部结构 + 最小意义单位（morpheme）form ↔ meaning interface temporal + frontal ｜**Syntax**（句法）hierarchical structure（层级结构）recursion（递归）embedding（嵌套）≠ linear order Broca's area ｜**Semantics**（语义）（literal meaning）context-independent（相对）temporal ｜**Pragmatics**（语用）语言在具体语境中的真实意义 speaker intention + context frontal + right hemisphere ｜**Modular view**（模块论）各层级独立｜**Interactive view**（交互论）同时处理相互影响｜现代主流：strong interaction（强交互）｜语言处理结合 N400 parallel + predictive processing
Broca's area 位置：左额叶（inferior frontal gyrus）(-54,12,22) BA44/45 功能：句法构建（syntax）语言产生（production）控制与选择（control）本质：**“怎么说”** Broca: + localization (function ↔ region) + lateralization (left dominance) BUT: → not single area → brain = network → lateralization ≠ absolute **Wernicke's area** 位置：左颞叶（posterior superior temporal gyrus）(-53,-31,9) BA22 功能：语义处理（semantics）词汇理解（lexicon） 本质：**“说的是什么意思”** **Dual Stream Model**（双通路模型）核心思想：语言处理分为两条通路：声音 → 意义（腹侧） 和 声音 → 发音（背侧）总结构图（必会画） Auditory input ➡️Temporal cortex ➡️VENTRAL STREAM（理解）➡️Meaning ｜ ➡️ DORSAL STREAM（产生）➡️ Motor / speech 腹侧通路（**Ventral Stream**）路径：temporal lobe → semantic system 功能：声音 → 意义（**sound** → **meaning**）特点：语义理解 + 词汇加工 + 自动处理 损伤结果：Wernicke’s aphasia + comprehension deficit + N400 异常 背侧通路（**Dorsal Stream**）路径：temporal → parietal → frontal（Broca）功能：声音 → 发音（**sound** → **articulation**） 特点：语音映射 + 句法处理 + 工作记忆参与 损伤结果：Broca’s aphasia + agrammatism + speech production difficulty 版本**1**（基础）The dual stream model proposes that language processing involves a ventral stream for comprehension

and a dorsal stream for speech production. 版本**2**（高分）Language processing relies on a distributed network in which the temporal lobe supports semantic representation while the frontal lobe supports syntactic and motor planning, connected via ventral and dorsal pathways.
BCI（脑机接口）BCI 依赖已有 brain mapping（如 temporal–frontal network）Neural activity ➡️Signal decoding ➡️Pattern recognition➡️Speech / text output（signal complexity）高维 噪声大 个体差异强 EEG 空间差 fMRI 时间慢 invasive electrodes 风险高 ｜individual variability 脑结构不同 激活模式不同 模型难以泛化（generalization problem）｜neural signal ≠ language meaning 大脑没有“单一编码方式”对应单词或句子｜可塑性（plasticity）大脑会变化：学习 重组 模型会失效（drift problem）｜更适用：严重运动障碍患者 ALS locked-in syndrome ｜优势：cognition intact, motor output impaired｜不适用：认知/语言系统受损者 Wernicke aphasia, severe comprehension deficit 输入本身已经错 → 无法正确解码, network disrupted 问题：no stable mapping 普通人（短期）原因：成本高 invasiveness 训练成本
brain methods：fMRI 血氧变化（BOLD）间接 neural activity｜EEG / ERP 电信号 时间动态 ｜Lesion 功能缺失 因果关系 ｜activation = 相对基线的统计变化（relative increase in signal）是 correlation（相关）不是 causation（因果）一个区域“激活”可能表示：执行任务 ✔️支持任务 ✔️抑制其他过程 ✔️或只是共现 ❗Activation does not imply necessity. ｜Region vs Pathway 灰质（cortical regions）处理信息 白质（white-matter pathways）连接信息 EEG measures changes in electrical voltage at the scalp, reflecting synchronised activity from large groups of cortical neurons ｜fMRI measures the BOLD signal: changes in blood oxygenation/blood flow that track local neural workload(an indirect proxy)｜TMS delivers a brief magnetic pulse that induces an electric field in the cortex, allowing us to disrupt processing in targeted regions
neuronal parts1️⃣Dendrites（树突）接收信号 输入整合（input integration）来自多个神经元 不同强度（excitatory / inhibitory）不是简单接收，而是加权整合（weighted summation）2️⃣Axon（轴突）发送信息 输出通道（signal propagation）action potential（动作电位）all-or-none（全或无）信息编码不是“强度”，而是“发放频率（rate coding）3️⃣Synapse（突触）神经元之间的连接 信息转换 + 调制（modulation）electrical → chemical → electrical neurotransmitters synapse 是“学习发生的地方”（synaptic plasticity）Neural computation arises from the integration of inputs at dendrites, propagation along axons, and modulation at synapses, forming the basis of higher-level cognitive processes such as language. ｜
Cognition = computation（grey matter） + connectivity（white matter）Grey Matter：cell bodies dendrites + synapses + 构成大脑皮层（cerebral cortex）highly folded 增加表面积 更强计算能力｜White Matter：axons+bundled into tracts 白质损伤：arcuate fasciculus→ conduction aphasia 即使区域正常，连接断裂也会导致语言障碍 Function is not only localized in regions, but distributed across connected networks
Brain Structure: Corpus Callosum: interhemispheric integration, enables lateralization to work as a coordinated system 损伤后 split-brain / callosal lesion 左右信息无法共享 |Arcuate Fasciculus: link B + W : 语音信息传输+repetition（重复）+语音工作记忆| Orientation: ⬆️dorsal/superior ⬇️ventral/inferior ⬅️rostral/anterior ➡️caudal/posterior| the x portion of x | Horizontal(axial/transverse) 顶上往下看Medial–Lateral view, Coronal 从背后看, Sagittal 从侧面看
Wada test: Language lateralization can be experimentally tested by temporarily disrupting one hemisphere. Support causation. L hemisphere: syntax + phonology + lexical RH: prosody + emotion + discourse Wada test 支持“lateralization”，但不支持“localization” 存在 variability（个体差异）Dominance does not imply exclusivity 问题：侵入性（invasive）风险高 粗粒度（coarse）不完全封闭半球 partial suppression + spillover effects, unnatural process 现代替代方法fMRI lateralization TMS The Wada test demonstrates hemispheric dominance by temporarily disrupting one hemisphere, revealing that language is typically left-lateralized but not exclusively localized.
Language is typically left-lateralized, with around 96% of right-handers and 60% of left-handers showing left-hemisphere dominance, but the reasons for this pattern remain multifactorial rather than singular. One explanation is the motor control hypothesis, which suggests that language emerged from left-hemisphere specializations for fine motor sequencing, particularly in regions such as Broca's area; however, this account mainly explains production and not comprehension. A second explanation is the processing efficiency hypothesis, which argues that concentrating language within one hemisphere reduces the need for interhemispheric transfer via structures like the Corpus Callosum, thereby enabling faster and more efficient computation, though this does not explain why the left hemisphere is favored. A third account emphasizes innate biological asymmetries, proposing that the brain is predisposed toward left-hemisphere language processing early in development, yet this cannot fully explain observed individual variability. Importantly, lateralization is probabilistic rather than deterministic, as evidenced by variation across individuals, particularly among left-handers. More broadly, functional differences between hemispheres may also play a role, with the left hemisphere supporting sequential and rule-based processing essential for syntax, and the right contributing to prosody and discourse. Overall, language lateralization likely arises from the interaction of

biological predisposition, motor specialization, and processing efficiency, and reflects dominance rather than exclusivity within a distributed, bilateral network.
Domain-General 语言依赖通用认知系统，而不是专门语言系统 attention+working memory+cognitive control｜Prefrontal cortex｜句子理解困难 ≠ 语法问题 = working memory overload **Domain-Specific** 存在专门用于语言的神经系统 phonology+lexical access+syntax ｜Broca's area+temporal language areas｜Language relies on domain-specific representations and domain-general control processes
Lesion - natural causal - necessity | fMRI - correlated - involvement | EEG - active changing - time|TMS -experimental causal(temporal) - necessity | stoke 局部、突然, tumor 渐进、可塑性强, neurodegenerative 广泛, TBI 不规则
UG:+ innate system+ explains acquisition Strong → domain-specific Minimal → hybrid → debated → not localized
Is language processing left-lateralized in newborns? Stimuli 语言刺激（如语音 syllables：ba / da）speech vs non-speech Step 1：建立 baseline 测 sucking rate Step 2：dichotic presentation 同时播放：左耳：刺激 A 右耳：刺激 B Step 3：测量 sucking rate: sucking frequency+sucking amplitude|Right ear > Left ear so left hemisphere dominance | Apparent inconsistencies in lateralization findings reflect methodological and developmental differences, as well as the multi-level and dynamic nature of language processing: age (development) + state (awake/sleep)+ method (EEG vs fMRI)+ stimuli (phonology vs syntax)+ baseline
TMS magnetic field → induced current → neural disruption(not activation) TMS = reversible/temporal visual lesion, but small effects, not precise, network impact
Structural: VBM links brain structure (grey matter density/volume) with experience and behavior, but does not establish causality. 学习新语言 → 灰质是否增加？experience-dependent plasticity（经验依赖的可塑性）某区域灰质密度 ↔ 语言能力structure–behavior relationship 测量对象：grey matter density grey matter volume 来自MRI 对应neural tissue properties（结构，而非活动）association not causation **DTI** links brain connectivity (white-matter pathways) with behavior and learning, but remains correlational.测量水分子扩散（diffusion）可以推断：axon bundles,white matter tracts 关键指标：FA（fractional anisotropy）反映连接“完整性/方向性” 回答群体差异（Group differences）experience → connectivity differences长期训练 → 白质改变，学习变化（Training effects）plasticity in connectivity ｜DTI = studying how brain regions communicate｜brain = nodes (VBM) + connections (DTI) + activity (fMRI) | but: indirect, correlational, ambiguous tracts
Functional: PET measures metabolic activity as an indirect proxy of neural function, offering molecular specificity but limited temporal precision. neural activity ↑→ energy demand ↑→ glucose use ↑ / blood flow ↑→ PET signal ↑ | PET provides molecular-level information about brain function(fMRI can’t do) 空间分辨率（中等）比 EEG 好略差于 fMRI|But 时间分辨率极低 无法测：快速语言 processing, ERP（如 N400）,放射性（严重限制）,间接测量和 fMRI 一样：proxy（代理信号）**fMRI** measures hemodynamic responses as an indirect proxy of neural activity, providing high spatial but low temporal resolution. neural activity ↑→ energy demand ↑→ blood flow ↑→ oxygenated hemoglobin ↑→ BOLD signal ↑| 高空间分辨率 可以精确定位B, 非侵入性,无辐射,可重复实验, whole-brain coverage 可以同时看：多个区域 network |But 间接测量血流 ≠ 神经可能：excitatory,inhibitory 时间分辨率差 不适合：N400,语言时间过程, 对运动敏感（motion artifact）, 噪声大（scanner noise）,correlation **fNIRS** provides a practical, **child-friendly** alternative to fMRI for studying cortical hemodynamic responses, at the cost of depth and spatial precision. 测血氧（hemodynamic response）可移动（portable）,对运动不敏感（tolerant）儿童友好（child-friendly）无噪声无密闭空间,|But 深度限制测不到：深层结构（如 hippocampus）空间分辨率较差<fMRI,时间分辨率仍然差也是 hemodynamic response
ERP allows precise tracking of the time course of cognitive processes, despite limited spatial localization. EEG 连续记录脑电信号 ERP 将 EEG 信号与事件（stimulus）对齐并平均 stimulus → brain response → average → ERP waveform 毫秒级时间分辨率 N170（视觉）N400（语义）But EEG spatial resolution 差 结果：只能说：“frontal”通过建模：推测来源 N170 → occipitotemporal 仍然是估计（inference），不是直接测量 | N400语义整合 P600 句法处理
MEG combines high temporal resolution with improved source localization, offering a bridge between EEG and fMRI. neural current → magnetic field → sensors detect → brain activity 神经活动产生：电信号（EEG测）磁信号（MEG测）毫秒级时间分辨率, 更好的空间定位（比 EEG）更好的 source localization| BUT 非常昂贵, 对脑区不均匀敏感 MEG 更容易检测：cortical surface 特定方向的电流 不容易检测：深层结构 某些方向 not equally sensitive to all sources 仍然存在定位问题 仍需：建模 推断 MEG = 时间 + 更好的空间（但仍不是完美）
Time (when) → EEG/ERP / MEG + Space (where) → fMRI / fNIRS + Structure (brain changes) → VBM / DTI + Causality (necessary?) → TMS / lesion + Child-friendly → fNIRS
Subtraction assumes “pure insertion,” which is often violated in language tasks.1 Words vs Pseudowords 差异：lexical access（词汇提取）visual + orthographic + phonological + lexical,| Word vs Pseudoword: mostly lexical 核心问

题:1Pure insertion assumption 实际:多个过程同时改变2interaction problem processes 不是独立 所以:subtraction 不干净 **Parametric design** overcomes the limitations of subtraction by avoiding binary contrasts and capturing graded neural responses. Character vs pseudo-character → orthographic processing 仍然依赖“pure insertion”假设 可能问题:semantic contamination, visual differences 核心思想 操控一个连续变量(continuous dimension)1Orthographic legality(正字法合法性)illegal → pseudo → real character2Visual complexity(视觉复杂度)low strokes → high strokes3Character frequency(字频)高频 → 低频 **Parametric design reveals sensitivity gradients rather than binary differences(subtraction) Factorial design** allows researchers to examine the independent and interactive effects of multiple factors on behavior or brain activity. It enables testing not only whether factors matter, but whether they interact. Interaction(交互作用): 一个因素的效果取决于另一个因素 2X2
实验设计(2 × 2 factorial)Factor 1:Frequency(词频)Factor 2:Regularity(规则性)行为基础(RT pattern)
Low-frequency irregular → slowest + High-frequency irregular → faster + Regular → less affected 脑成像设计(fMRI)
预测:Low-frequency irregular → 最大 activation 因为:retrieval difficulty ↑processing cost ↑ Irregular → lexicon
Regular → rule system
Blocked designs maximize statistical power, while **event-related designs** maximize experimental flexibility and temporal specificity. 同一条件连续呈现(block)1信号叠加(signal summation)信号更强 → 更容易检测2噪声减少| BUT1habituation(习惯化)大脑:逐渐适应 信号下降2设计不灵活不能:随机化 trial-level 分析 Event-related Design 单个 trial 随机呈现1高灵活性 随机化 多条件 trial-by-trial2避免 habituation 刺激变化3可以分离过程 不同事件 | BUT1信号弱 没有累积2统计 power 低 更难 detect 小效应
Sparse scanning 利用 fMRI BOLD 信号的时间延迟，在“静音窗口”呈现听觉刺激，从而避免 scanner noise 对听觉任务的干扰。Sparse sampling separates stimulus presentation from image acquisition, enabling cleaner measurement of auditory processing in fMRI. +1quieter sequences2noise attenuation(耳机等)3modeling noise in analysis
The brain transforms continuous acoustic signals into stable speech-relevant categories through a combination of perceptual tuning, neural encoding, and categorical decision processes. Early auditory regions (e.g., primary auditory cortex) encode fine-grained acoustic features such as frequency, timing, and amplitude in a largely continuous manner. As processing progresses along the auditory pathways—particularly into superior temporal regions—these continuous inputs are mapped onto learned phonetic categories based on experience with a specific language. This involves **categorical perception**, where sensitivity to within-category variation is reduced while sensitivity to between-category differences is enhanced. Neural populations become selectively tuned to category boundaries (e.g., voice onset time for /b/ vs /p/, or pitch contours for tones), effectively compressing variability within a category and exaggerating differences across categories. Higher-level regions, including frontal areas, contribute top-down influences such as attention, prediction, and context, helping stabilize perception under noisy or ambiguous conditions. Through this hierarchical and interactive process, the brain converts variable acoustic input into discrete, language-relevant representations that support efficient recognition and comprehension. **categorical perception: + efficiency - flexibility** Categorical perception leads to perceptual narrowing, enhancing native-language processing at the cost of reduced sensitivity to non-native phonetic distinctions.
auditory pathway:Cochlea → brainstem → IC → MGN → cortex ｜→ transduction → integration → abstraction
Primary auditory cortex(A1)Heschl's gyrus BA 41，A1 处理“声音是什么”，而不是“声音代表什么” 频率(pitch)强度(loudness)时间结构 Association auditory areas(BA 42)区域belt+parabelt1特征整合(feature integration)2类别选择性(category selectivity)区分:speech3连接意义与动作 ｜ **A1**提供输入 ➡️**Association cortex**:speech perception + phoneme + word 进一步连接:temporal lobe(语义)+frontal(控制)
Word > Tone → language + semantics + Tone > Noise → structure ｜ → hierarchy BUT: ≠ lateralization ≠ pure language The **subtraction** results suggest a hierarchical organization of auditory processing, but they cannot be used to directly infer language lateralization, as the contrasts include multiple overlapping processes.
Has the learner acquired new phonetic contrasts in English?1Identification task(分类任务)听声音 → 判断:“r” or “l”2Discrimination task(区分任务)判断:是否相同 ｜continuum(连续刺激)The /r–l/ contrast is ideal because it is not phonemically distinguished in Japanese, making it a sensitive test of perceptual learning.1分类曲线(identification curve)如果学会:曲线更陡(categorical boundary)2discrimination peak 边界处:区分能力 ↑
Motor Theory:1Double Dissociation(最强反证)Double dissociations demonstrate that speech perception and production rely on partially independent systems.2婴儿证据(developmental evidence)perception can exist without motor knowledge3替代理论存在(不是唯一解释)Auditory and perceptual accounts can explain categorical

perception without invoking motor representations. **MVPA** shifts the focus from “how much activity” to “what information is represented.” Univariate → strength + MVPA → pattern → decode information → distributed coding
Neural activation in this context is likely more similar to **speech perception** than speech production, as it primarily involves processing auditory or linguistic input rather than generating articulatory output. However, it is not purely perceptual: evidence suggests that speech perception engages not only auditory regions but also, to some extent, motor-related areas, reflecting a partially shared network. In this sense, the activation may also resemble **imagining someone else speaking**, as both involve internally simulating speech without overt articulation. Such imagery recruits auditory representations and can engage predictive and motor systems, but without the full motor execution required for production. Tf, the neural pattern is best characterized as primarily perceptual with possible motor simulation components, consistent with hybrid models of speech processing that integrate auditory + motor contributions.
According to the dual-stream model, speech perception involves a **ventral stream** for mapping sound to meaning and a **dorsal stream** for sensorimotor integration. In a **listening comprehension task while eating**, processing is expected to rely primarily on the ventral stream, as comprehension of familiar language does not require articulatory mapping; thus, concurrent oral motor activity (eating) should have minimal interference. In contrast, when learning a new language with unfamiliar phonetic contrasts, the dorsal stream becomes more engaged, as learners rely on sensorimotor mapping and articulatory simulation to establish new phonetic categories. In this case, eating may interfere with performance by occupying the articulatory system, reducing the efficiency of dorsal stream processing. Therefore, the model predicts little interference in comprehension of familiar language, but greater interference when learning novel phonetic contrasts due to increased reliance on the dorsal pathway.
MTG/STG（BA21/22）→ activation + pMTG/AG（BA39）→ semantics + Ventral IFG → competition + Dorsal IFG → selection 0 ms → sound onset 250–300 ms → **PMN (pre-lexical) 400–600 ms → N400 (lexical/semantic)** ERP evidence suggests a temporal progression from phonological mapping (PMN) to lexical-semantic integration (N400), indicating that speech comprehension unfolds in stages. In speech processing, **cohorts（BA 21/22）**(e.g., *road* for *rose*) and **neighbours** (e.g., *rope*) show different temporal patterns across ERP components. At the early **PMN stage (≈250–300 ms)**, which reflects pre-lexical phonological mapping, cohorts are processed similarly to exact matches because they share the same initial sounds and are activated as part of the initial candidate set, whereas neighbours, which do not fully match the onset, already show greater mismatch responses. However, at the later **N400 stage (≈400–600 ms)**, which reflects lexical-semantic processing and competition resolution, both cohorts and neighbours diverge from the exact match, as the system begins to eliminate incorrect candidates; this results in increased N400 amplitudes for both, though for different reasons (cohorts are rejected after initial activation, while neighbours were weaker candidates from the start). This pattern demonstrates that speech recognition is a dynamic process, where early stages are driven by phonological overlap and later stages by lexical competition and selection.
The mental lexicon is not localized to a single brain region but is instead represented within a distributed neural network. Core components of this network include a predominantly left-lateralized temporal system (e.g., Superior Temporal Gyrus and Middle Temporal Gyrus), which supports online lexical access and the mapping of speech sounds onto meaning. The Anterior Temporal Lobe functions as a transmodal conceptual hub, integrating semantic information across different modalities (“spokes”). In addition, frontal regions such as the Inferior Frontal Gyrus are involved in control processes, including lexical selection, competition resolution, and retrieval under ambiguity. Together, these regions form a distributed and interactive system in which lexical knowledge is both represented and dynamically accessed during language processing. ｜**Neural evidence supporting the dual-route model** would show **distinct and dissociable pathways** for visual word processing, analogous to the ventral (sound-to-meaning) and dorsal (sensorimotor/phonological mapping) streams in speech perception. Specifically, we would predict a **ventral occipito-temporal–temporal pathway** (including the Visual Word Form Area and temporal semantic regions) to be more engaged for **familiar words and irregular words** that rely on direct lexical-semantic access, and a **dorsal parietal–frontal pathway** (including inferior parietal regions and Inferior Frontal Gyrus) to be more engaged for **pseudowords and low-frequency regular words** that require grapheme-to-phoneme conversion. Appropriate contrasts would include **words vs pseudowords** (to isolate lexical access), **irregular vs regular words** (to test reliance on lexical vs rule-based routes), and critically **interactions such as (pseudowords > words) and (irregular > regular)**. Strong support would come from **double dissociations**, where ventral regions preferentially respond to lexical/semantic processing and dorsal regions to phonological/assembly processing, demonstrating that reading relies on two functionally distinct but interacting neural routes. ｜ **Reading aloud** is supported by multiple interacting

pathways, primarily a **sublexical pathway** and a **lexical-semantic pathway**, whose relative contribution depends on task demands and reader experience. The sublexical pathway, associated with dorsal fronto-parietal regions (e.g., Inferior Frontal Gyrus and parietal cortex), enables grapheme-to-phoneme conversion and is especially important for unfamiliar letter strings and pseudowords. In contrast, the lexical-semantic pathway, involving ventral occipito-temporal regions such as the Visual Word Form Area and temporal semantic areas, supports rapid access to stored word forms and meanings, and is crucial for familiar and irregular words that cannot be decoded via rules. Importantly, these pathways are not used in isolation; rather, their weighting varies with word type, writing system, reading strategy, and expertise—for example, beginning readers rely more heavily on the sublexical route, while skilled readers increasingly utilize the lexical-semantic pathway for efficient recognition. ｜**Word meaning** can, to a large extent, be assessed **without explicitly evaluating pronunciation**, as evidence suggests that semantic access can proceed via a direct lexical-semantic pathway without obligatory phonological mediation. However, phonological information is often **automatically activated** during reading, even when it is not required by the task, reflecting the tight coupling between orthography and phonology in skilled readers. Crucially, automatic activation does not imply necessity: studies using tasks such as semantic judgment and masked priming show that meaning can still be accessed when phonological processing is disrupted or minimized, indicating that phonology is not strictly required. Therefore, while phonological processing is typically co-activated and may facilitate or constrain lexical access, it is **not necessary for accessing word meaning**, supporting dual-route models in which orthography can map directly onto semantics.
The triangle model proposes that **reading** arises from learned, interactive mappings among **orthography (O), phonology (P), and semantics (S)**, rather than from separate, independent routes. In this framework, multiple pathways operate **in parallel**: the O→P pathway supports grapheme-to-phoneme decoding, especially for unfamiliar words and pseudowords; the O→S pathway enables rapid access to meaning for familiar words; and the S→P pathway helps stabilize pronunciation when spelling–sound mappings are inconsistent, as in irregular words. Unlike dual-route models, which posit distinct lexical and sublexical routes, the triangle model emphasizes a **distributed, graded system** in which processing emerges from the interaction of all three components, with their relative contributions depending on factors such as word familiarity, regularity, and reading experience.
The brain is not evolutionarily specialized for reading or print in the same way it is for spoken language, as writing systems are too recent in human history to have dedicated, innate neural structures. Instead, reading relies on a process of **neuronal recycling**, whereby pre-existing visual and language networks are repurposed for written language processing. A key region in this network is the Visual Word Form Area, which becomes selectively responsive to written words through experience. However, this region is not exclusively dedicated to reading; rather, it emerges from visual object recognition systems and interacts with temporal language areas and frontal control regions. Therefore, while the brain develops **functional specialization for print with experience**, this specialization is learned and distributed, not innate or strictly modular. ｜The left-lateralization and role of the **Visual Word Form Area** in reading can be explained by two complementary accounts. The **language account** argues that the VWFA becomes specialized for reading due to its anatomical location within the left hemisphere, where it is optimally positioned between visual processing regions and anterior language areas, allowing efficient connectivity for mapping visual input onto phonological and semantic representations. In contrast, the **visual perceptual account** proposes that the VWFA is not inherently specialized for reading, but is recruited for this function because it is well-suited for processing visually complex, high spatial frequency stimuli (such as written words) in parallel. Together, these accounts suggest that VWFA specialization emerges from an interaction between **connectivity constraints (language network)** and **intrinsic visual processing biases**, consistent with the idea of neuronal recycling, where pre-existing visual systems are repurposed for reading. **Individually-defined Visual Word Form Area** is necessary because its functional location varies across individuals and forms a gradient-like patch rather than a fixed anatomical point. Using group-defined ROIs can blur or mislocalize the true VWFA signal, as averaging across participants may mix a participant's VWFA with neighboring non-word-selective regions, thereby reducing sensitivity and obscuring effects. This issue is compounded by the fact that different localizer contrasts (e.g., words > consonant strings vs. words > numbers) can yield slightly different peak activations, meaning there is no single canonical coordinate for VWFA across studies. Therefore, defining VWFA individually using a functional localizer ensures that analyses target the correct, participant-specific region, improving both sensitivity and interpretability of results.
The representation of two languages in **bilingual brains** is best understood as a **partially overlapping and dynamic system**, rather than strictly shared or fully separate. Neuroimaging evidence shows that L1 and L2 largely recruit

overlapping regions within the left-lateralized language network, particularly in temporal and frontal areas, supporting the **shared representation view**. However, differences can emerge depending on factors such as age of acquisition, proficiency, and context of use, with L2 sometimes showing additional or shifted activation, reflecting greater processing demands or less efficient representations. Thus, while the core neural substrate is largely shared, bilingual language representation is **modulated by experience**, resulting in graded overlap rather than strict separation.
Assimilation and accommodation represent two complementary mechanisms underlying second language (L2) processing in the brain. **Assimilation** refers to the use of existing L1 neural networks to process L2, leading to largely overlapping activation patterns, especially in highly proficient or early bilinguals where L2 can be efficiently integrated into established language systems. In contrast, **accommodation** involves the recruitment or adaptation of additional neural resources to handle linguistic features that are absent or difficult in L1, such as novel phonetic contrasts or grammatical structures, often observed in late or less proficient learners. Neuroimaging evidence suggests that bilingual language representation reflects a **dynamic balance between assimilation and accommodation**, with the extent of each depending on factors such as age of acquisition, proficiency, and linguistic distance between the two.
The longer processing time observed for dual-language sentences does not necessarily reflect a language-specific **switch cost**, nor does it imply that bilinguals access their languages independently. Instead, such effects can be explained by **domain-general control demands**, including increased attentional load, conflict monitoring, and selection between competing alternatives. A switch cost does not inherently support language-specific access, because it may arise from the need to resolve competition when both languages are simultaneously activated. In fact, the same findings are compatible with a **language-independent access account**, in which both languages are co-activated and the observed delay reflects the effort required to suppress or select among competing representations. Therefore, switch costs are better interpreted as evidence of **cognitive control mechanisms operating over a shared bilingual system**, rather than proof of separate language-specific processing routes.
Whether it is easier to manage two similar languages (e.g., Italian–Spanish) or two distinct languages (e.g., English–Chinese) depends on the type of cognitive demands involved, as both situations impose different forms of difficulty. Similar languages tend to create **greater lexical and phonological competition**, as overlapping forms and meanings increase co-activation and require stronger inhibitory control to suppress non-target representations. In contrast, distant languages reduce direct competition but impose higher demands on **monitoring and switching**, as speakers must manage distinct phonological systems, scripts, and grammatical structures. Therefore, similar languages are more demanding in terms of **competition resolution**, whereas distant languages are more demanding in terms of **control and switching across systems**, suggesting that bilingual language control involves a trade-off rather than a single dimension of difficulty. **The neural and cognitive advantages associated with bilingualism** are unlikely to be unique, as similar benefits can arise from other experiences that place sustained demands on cognitive control, attention, and learning. Activities such as musical training, video gaming, and intensive skill learning also engage executive functions and can lead to comparable enhancements in brain structure and function. From this perspective, bilingualism may be one instance of a broader class of experiences that strengthen domain-general control systems rather than a special case. Furthermore, evidence from short-term language training suggests that even relatively brief exposure can induce measurable changes in brain structure and function, indicating that the brain remains highly plastic and responsive to learning. Therefore, bilingual advantages are better understood as part of a continuum of experience-dependent neurocognitive adaptation, rather than as a unique property of bilingualism itself.
Bilingualism is associated with a range of neural consequences that reflect experience-dependent plasticity across both functional and structural levels of the brain. Functionally, bilinguals often show increased engagement and efficiency in control-related regions such as the Inferior Frontal Gyrus and Anterior Cingulate Cortex, which support language selection and conflict resolution between co-activated languages. Structurally, studies using techniques such as VBM and DTI have reported changes in grey matter density and white matter connectivity, particularly in language and control networks. Additionally, bilingualism can strengthen connectivity within fronto-temporal networks and may contribute to neural reserve, potentially enhancing resilience against cognitive decline. Importantly, these effects are not uniform but depend on factors such as proficiency, age of acquisition, and language use, highlighting that bilingualism shapes the brain in a graded and experience-dependent manner.
Classical aphasias reveal that language is **partly localized yet fundamentally distributed and functionally specialized across interacting networks**. Evidence from **Broca**'s area shows that damage leads to non-fluent, effortful speech with relatively preserved comprehension, indicating a critical role in speech production and syntactic

planning. In contrast, lesions to **Wernicke**'s area result in fluent but often meaningless speech with impaired comprehension, highlighting specialization for lexical-semantic processing. The presence of **double dissociations** between production and comprehension demonstrates that these functions rely on partially independent systems. At the same time, conduction aphasia and other syndromes implicate white-matter connections (e.g., pathways linking frontal and temporal regions), showing that **connectivity between regions is essential**. Overall, classical aphasias support a model in which language is organized into **specialized components (production, comprehension, repetition) distributed across a network**, rather than a single unified module.

I（**Input**）：听觉输入 **A**（**Auditory**）：词听觉中心（Wernicke）**C**（**Concept**）：概念/语义中心 **M**（**Motor**）：运动言语中心（Broca）**O**（**Output**）：发音输出 I → A → C → M → O ① **M** 损伤 → **Broca's aphasia** ❌ 说话 ✔️ 理解 ❌ 复述 ② **A** 损伤 → **Wernicke's aphasia** ✔️ 流利 ❌ 理解 ❌ 复述 ③ **A–M** 连接断裂 → **Conduction aphasia** ✔️ 说话✔️ 理解 ❌ 复述 ④ **M–C** 断裂 → **Transcortical Motor Aphasia** ❌ 自发说话 ✔️ 理解 ✔️ 复述 ⑤ **A–C** 断裂 → **Transcortical Sensory Aphasia** ✔️ 流利 ❌ 理解 ✔️ 复述 ⑥ **I–A** 断裂 → **Word Deafness**（纯词聋） ❌ 听懂语言✔️ 听到声音（非语言）✔️ 说话正常 ⑦ **M–O** 断裂 → **Anarthria / Motor output deficit** ❌ 发音 ✔️ 理解✔️ 内部语言正常

While the classical view assigned **language production to Broca's area and comprehension to Wernicke's area**, contemporary evidence suggests that this division is overly simplistic. Instead, language processing is now understood as a **distributed, interactive network** in which both regions contribute to multiple aspects of language. Broca's area is not limited to production but is involved in functions such as syntactic processing, cognitive control, and selection, while Wernicke's area is not solely responsible for comprehension but participates in mapping sound to meaning within a broader temporal network. Moreover, production and comprehension recruit overlapping neural systems, reflecting shared representations and processes. Thus, rather than discrete modules, Broca's and Wernicke's areas are best viewed as **key nodes within a larger language network**, each contributing specialized computations that interact dynamically during language use.

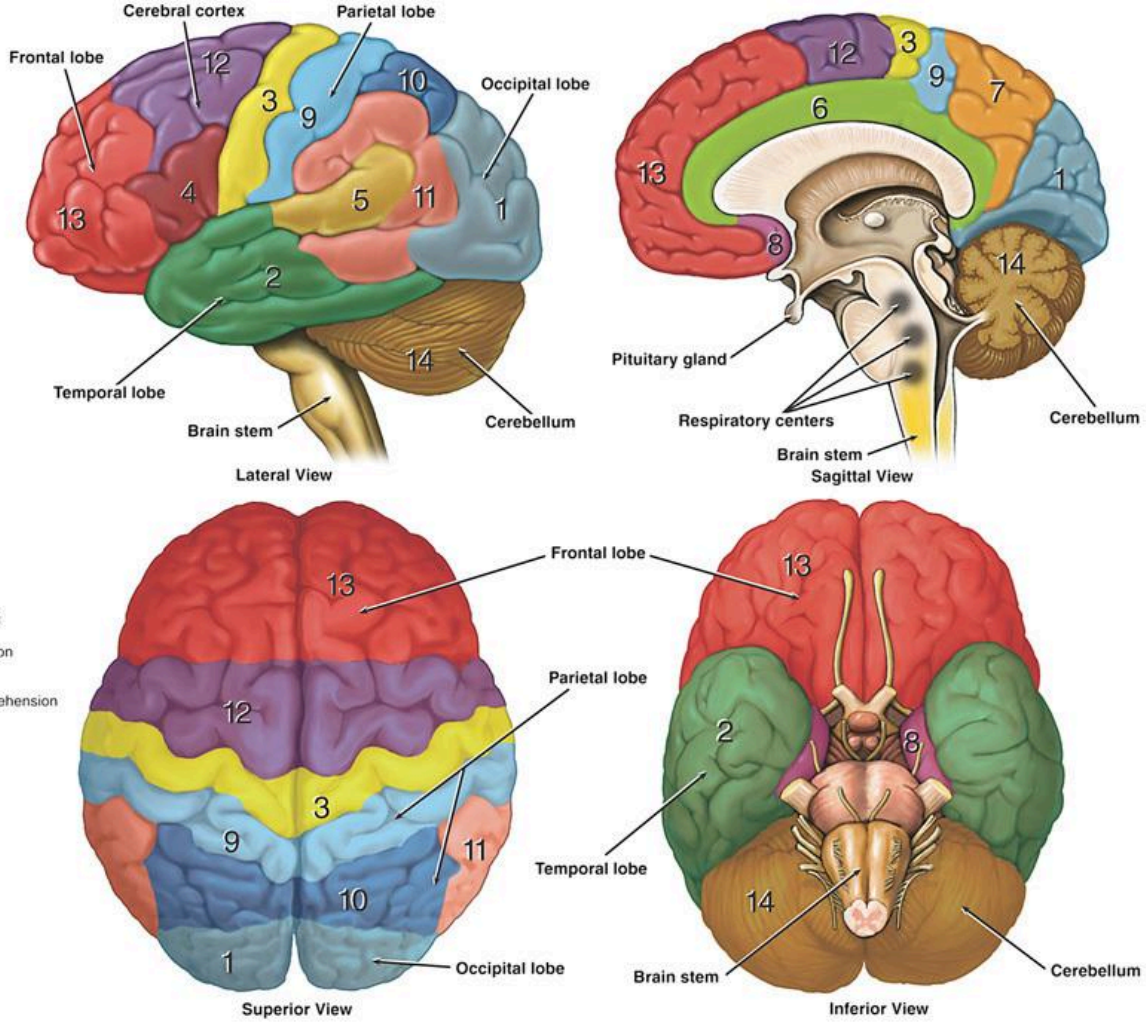
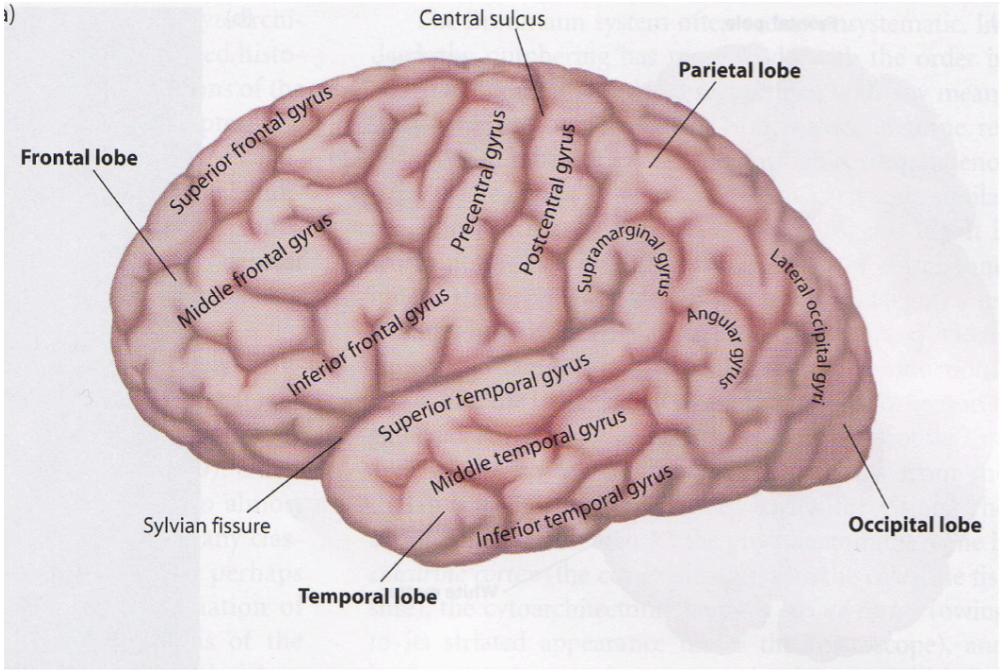
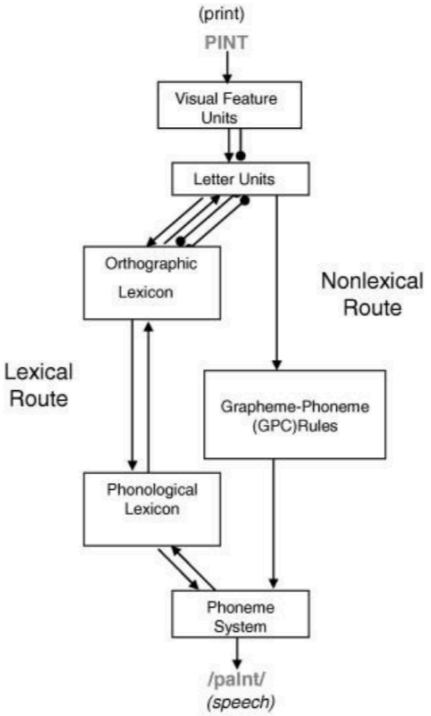
The neural basis of dyslexia is best understood as a disruption within a distributed reading network, particularly involving left-lateralized occipito-temporal, temporo-parietal, and frontal regions. Individuals with dyslexia typically show reduced activation in posterior regions, including the Visual Word Form Area and temporo-parietal areas associated with phonological processing, reflecting difficulties in mapping orthography to phonology. In contrast, increased activation is often observed in frontal regions such as the Inferior Frontal Gyrus, suggesting compensatory reliance on effortful, controlled processing. Structurally, differences in grey matter and white matter connectivity (e.g., weakened pathways linking posterior and anterior regions) have also been reported. Overall, dyslexia reflects a **network-level imbalance**, where inefficient phonological decoding and reduced posterior specialization lead to compensatory engagement of frontal control systems, rather than damage to a single localized area.

Aphasia	Fluency	Comprehension	Repetition	Location
Broca's	❌ 非流利	✔️ 较好	❌ 差	Cortical（前额叶 / Broca）
Wernicke's	✔️ 流利	❌ 差	❌ 差	Cortical（颞叶 / Wernicke）
Conduction	✔️ 流利	✔️ 好	❌ 差	Cortical（弓状束）
Transcortical Motor	❌ 非流利	✔️ 好	✔️ 好	Cortical（前额叶外侧）
Transcortical Sensory	✔️ 流利	❌ 差	✔️ 好	Cortical（颞顶交界）

Mixed Transcortical	❌ 非流利	❌ 差	✔️ 好	Cortical（广泛但保留连接）
Global	❌ 非流利	❌ 差	❌ 差	Cortical（广泛损伤）
Anomic	✔️ 流利	✔️ 好	✔️ 好	分布式（轻度）
Subcortical Motor	❌ 非流利	✔️ 好	✔️ 好	Subcortical（基底节）
Subcortical Sensory	✔️ 流利	❌ 差	✔️ 好	Subcortical（丘脑）
Thalamic Aphasia	✔️ 流利	❌ 轻-中度	✔️ 好	Subcortical（丘脑）
Pure Word Deafness	✔️ 流利	❌ 听不懂	✔️ 差	Cortical（听觉输入）
Apraxia of Speech	❌ 发音困难	✔️ 好	❌ 差	Cortical（运动规划）

Functional Areas of the Cerebral Cortex

1. **Visual Area:** Sight, Image recognition, Image perception
2. **Association Area** Short-term memory, Equilibrium, Emotion
3. **Motor Function Area** Initiation of voluntary muscles
4. **Broca's Area** Muscles of speech
5. **Auditory Area** Hearing
6. **Emotional Area** Pain, Hunger, "Fight or flight" response
7. **Sensory Association Area**
8. **Olfactory Area** Smelling
9. **Sensory Area** Sensation from muscles and skin
10. **Somatosensory Association Area** Evaluation of weight, texture, temperature, etc. for object recognition
11. **Wernicke's Area** Written and spoken language comprehension
12. **Motor Function Area** Eye movement and orientation
13. **Higher Mental Functions** Concentration, Planning, Judgment, Emotional expression, Creativity, Inhibition

Functional Areas of the Cerebellum

14. **Motor Functions** Coordination of movement, Balance and equilibrium, Posture